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INFORMATION TECHNOLOGIES

iPredicta

A look into the future of production assets

Information system of remote production assets monitoring, diagnostics and predictive analysis

http://iPredicta.ru

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Issues of the production assets management

- Substantial share of the old and imported assets
- Process of assets aging/shortage of highly qualified staff, knowledge and experience loosing
- Maintenance and Repairs costs cut demand
- Traditional diagnostics tools and methods do not allow to identify emerging issues and timely avoid accidents
- Lack of highly qualified resources at remote locations – demand of remote monitoring and control

Industrial companies challenges

- Operational costs cut
- The need to maintain the readiness of production infrastructure
- Usage of traditional Maintenance and Repairs approaches do not have a potential for future efficiency raising
- Substantial costs for accidents fixing

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The Asset Management System allows to increase the efficiency of production assets

- Improved financial results: ROI and cost reduction
- Making investment decisions based on objective information: balancing costs, risks, and prospects
- Managed risks: reducing financial losses, improving safety, health, negative impact on the environment, reputation
- Improved service and productivity: increased service/product quality, reduced downtime
- Demonstrating social responsibility: reducing emissions, saving resources
- Improved efficiency and effectiveness
- Improved sustainability of development
- Increased reputation

Source: GOST 55.0.01-2014

The monitoring, diagnostics and predictive analytics system is a key element of the Asset Management System linking Planning and Execution processes

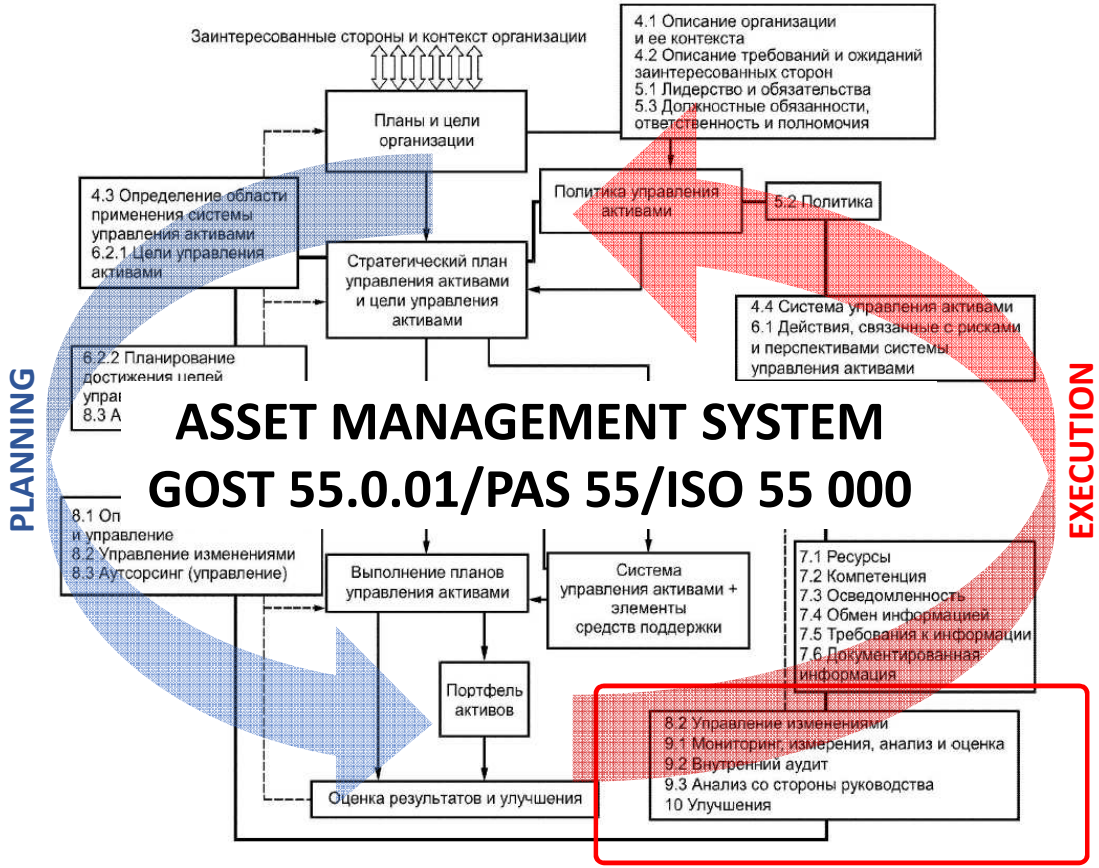

Figure 1. Relationship between key elements of the Asset Management System

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Potential impact of implementing Predictive Analytics system

- Reducing the likelihood of accidents, increasing the level of production safety
- Increase in equipment service life in the range of 20 to 40%
- Reduction in the number of equipment failures by an average of 30–35%
- Reduction in the volume of reactive (unscheduled) maintenance by 10–44%

Early detection of problems using Predictive Analytics helped avoid significant costs

- More than $4 million in costs were avoided through early detection of damage to rotating equipment (turbines, pumps)
- $370K saved by early warning of heat exchanger bypass valve problems
- Over $500K in costs saved by early detection of Impending clutch failure on factory engine
- $250K in annual savings achieved by identifying pump inefficiencies
- $250K in costs avoided with early warning of bearing seal pressure drop problem
- Over $243K in costs avoided by early detection of control valve mispositioning

Source: *Predictive Maintenance for Improved Performance in Oil and Gas, Kim Custeau, Director of Information and Asset Management, Schneider Electric*

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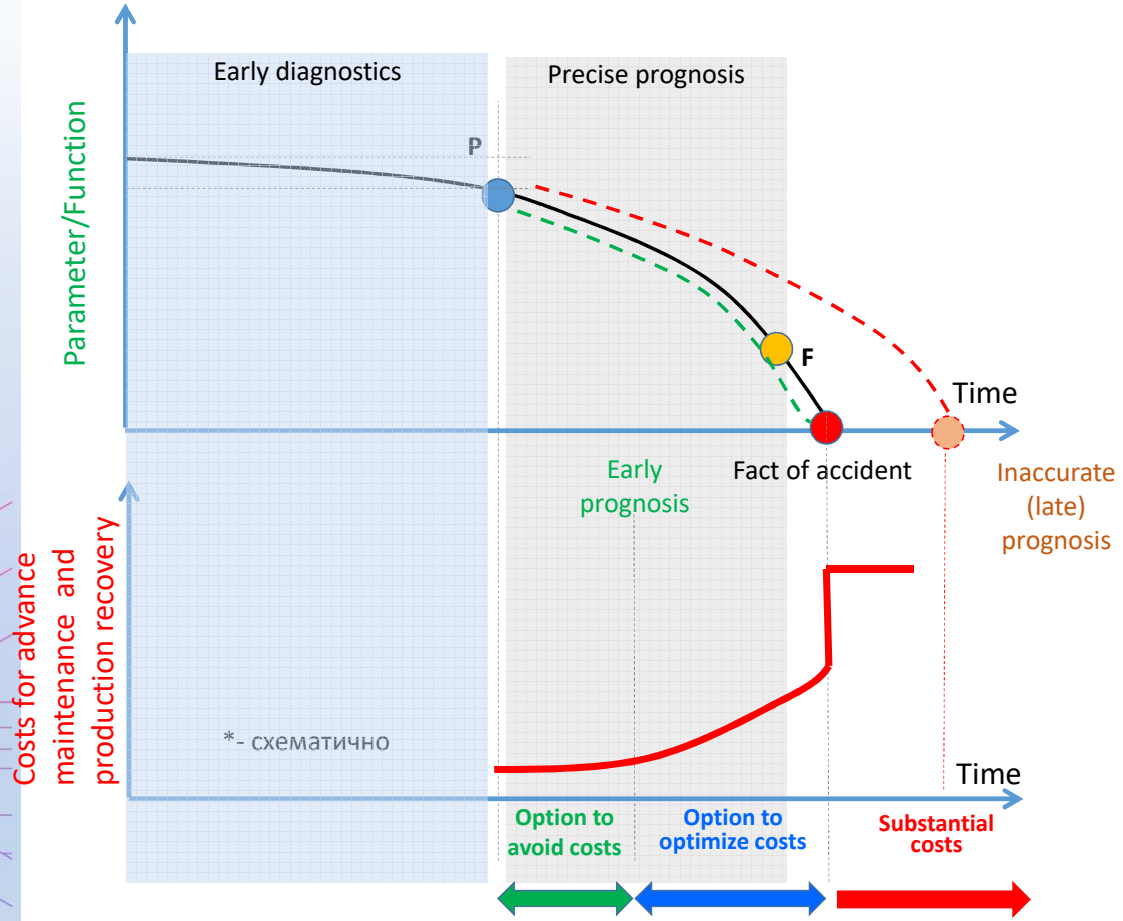
Main concepts in the iPredicta basis

- Diagnostics of technical status in real time mode, usage of “humanless production” model, cut of operational costs
- Early diagnostics of equipment degradation
- Maintenance and Repairs costs cut
- Precise calculation of remained equipment resource/ degradation parameters, focus on avoiding of accidents and on financial damage after an accident

Solution for Maintenance and Repairs costs optimization



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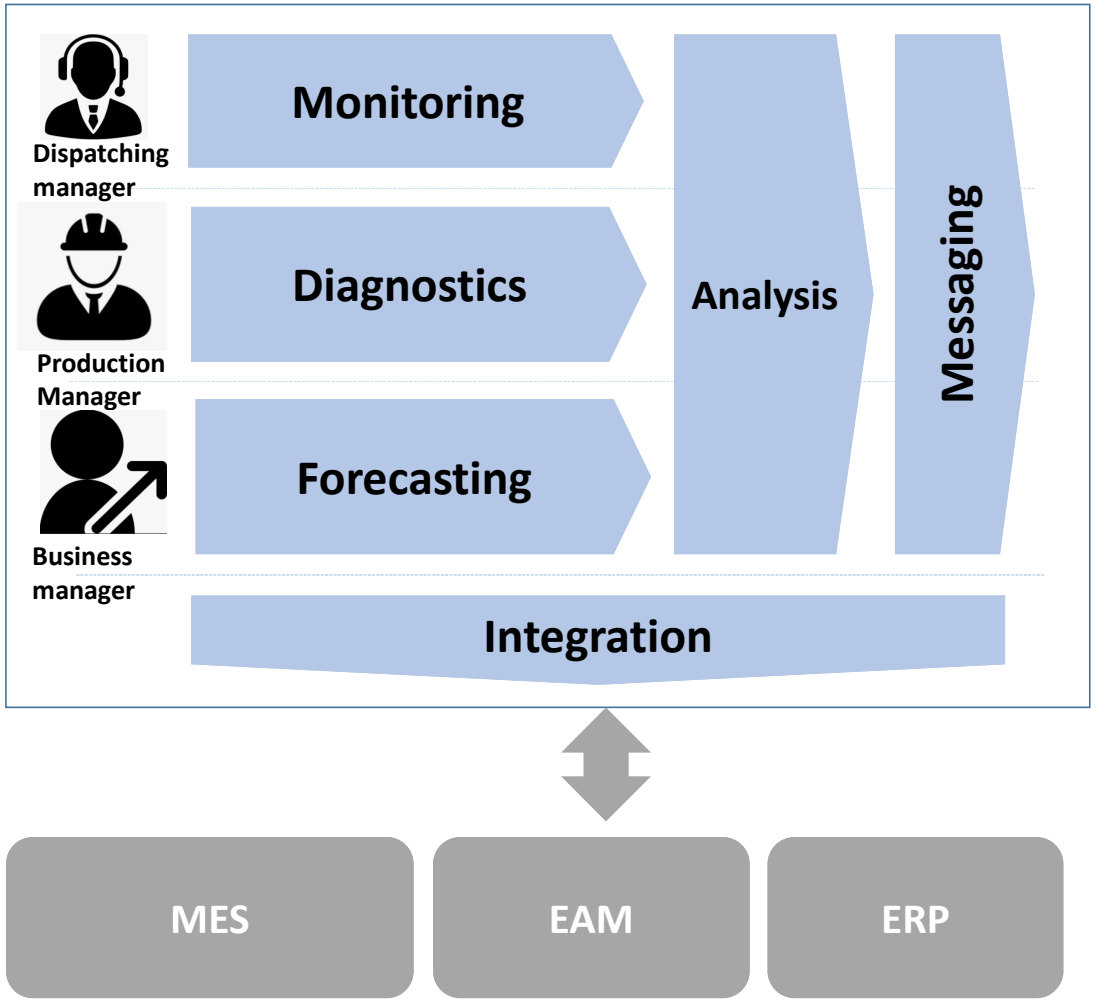
Business processes supported by the SW

- Full coverage of the end-to-end asset condition assessment process
- Reporting for production management and economical departments
- All kind of analytics



iPredicta

Key functionality

- Based on the Industrial Internet of Things (IIoT) platform DataPlat from InfTech
- Modern and effective diagnostic and forecasting algorithms
- Advanced analytics, modeling and reporting functions

Main functions of iPredicta

- Creation and maintenance of an information model of a “digital twin” of production facilities and equipment
- Calculation of indicators and characteristics of any complexity
- Monitoring and control of technical condition based on primary indicators and/or calculated characteristics
- Diagnostics of technical condition in real time using various methods (including machine learning) and state-of-the-art algorithms; proprietary algorithms for complex diagnostics
- Possibility of diagnosing both dynamic equipment (pumps, turbines, compressors) and static equipment (linear part, tanks, boilers)
- Diagnostics based on expert database
- Calculation of reliability indicators
- Forecasting technical condition indicators and characteristics in real time
- Automatic generation and management of monitoring and diagnostic events, generation of event notifications via various channel
- Analysis of the technical condition of an object based on historical and forecast indicators and characteristics. Modeling functions
- Flexible configuration of workflows for monitoring, diagnostics and forecasting
- Integration with external systems (ERP, APM, EAM, CMMS…)

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Key advantages

- iPredicta has broad functionality comparable with the best in class international technologies
- iPredicta – Russian development, not using licensed components

iPredicta applied technologies

АЛГОРИТМЫ

MSET

SPRT

HOTELLING T2

correlation

NEURAL

ТЕХНОЛОГИИ

INDUSTRY 4.0

IIOT

DIGITAL TWIN

Registry of Russian Software

Competitive advantages of iPredicta

SmartSignal | PdMS | PAA (PRiSM) | прана | Ctrl²go!

- Wide range of the diagnostic and predictive algorithms — Dinamic, Passive
- Algorithms of the complex diagnostics based several methods — 360°
- Integrated solution from one developer hands

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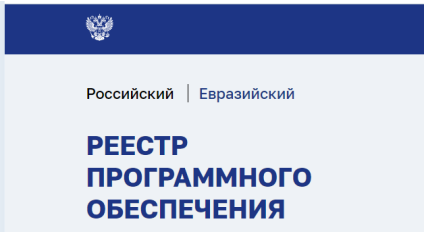
Intellectual property

- All SW components are registered in Russian IP rights authority and have proper certificates
- SW iPredicta is registered in the Ministry of digital development of Russia



Record in the Ministry of digital development of Russia

- Usage of the modern diagnostic tools, predictive analysis of the technical condition of equipment.
- Competitive advantages AVEVA Predictive Analytics, SAP PdMS, GE Smart Signal, Aspen NTEL.
- Can be applied for dynamical equipment (pumps, compressors, engines) as well for static equipment (pipes, electrical equipment, transform, boilers etc.)
- Deep integration in the main customer industrial business processes

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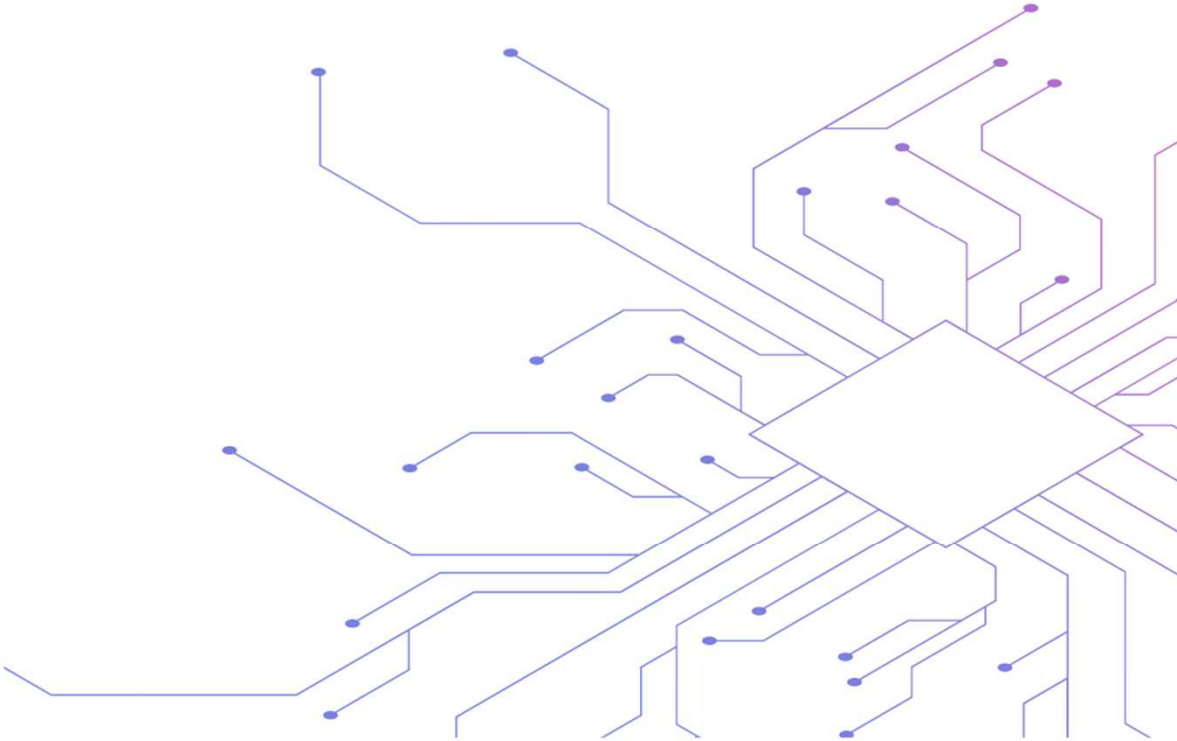
Digital technologies are the driver of economic development



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